

1. Solve a 3×3 linear system (by elimination)

Problem:

$$2x + y - z = 8$$

$$-3x - y + 2z = -11$$

$$-2x + y + 2z = -3$$

Steps (Gaussian elimination sketch):

- From eqn1: keep as is.
- Add 3×eqn1 to eqn2 $\rightarrow (-3+6)x + (-1+3)y + (2-3)z = -11 + 24 \rightarrow 3x + 2y - z = 13$.
- Add eqn1 to eqn3 $\rightarrow (-2+2)x + (1+1)y + (2-1)z = -3 + 8 \rightarrow 0x + 2y + z = 5$.
- From the last: $z = 5 - 2y$. Plug into the " $3x + 2y - z = 13$ ": $3x + 2y - (5 - 2y) = 13 \rightarrow 3x + 4y - 5 = 13 \rightarrow 3x + 4y = 18 \rightarrow x = (18 - 4y)/3$.
- Plug z and x into eqn1: $2 \cdot (18-4y)/3 + y - (5-2y) = 8 \rightarrow (36-8y)/3 + y - 5 + 2y = 8$. Multiply by 3: $36 - 8y + 3y - 15 + 6y = 24 \rightarrow 21 + y = 24 \rightarrow y = 3$.
- Then $z = 5 - 2 \cdot 3 = -1$; $x = (18 - 4 \cdot 3)/3 = (18 - 12)/3 = 2$.

Answer: $x = 2, y = 3, z = -1$.

2. Lagrange multipliers (quadratic with linear constraint)

Minimize $f(x,y) = x^2 + 2y^2$ subject to $g(x,y) = x + 2y - 6 = 0$.

Set gradients: $\nabla f = (2x, 4y) = \lambda \nabla g = \lambda(1, 2)$.

So $2x = \lambda$ and $4y = 2\lambda \rightarrow 2y = \lambda$. Hence $2x = 2y \rightarrow x = y$.

Constraint: $x + 2y = 6$ with $x = y \rightarrow y + 2y = 3y = 6 \rightarrow y = 2 \rightarrow x = 2$.

Check value: $f(2,2) = 4 + 8 = 12$.

Answer: Minimizer at $(x, y) = (2, 2)$, minimum value 12.

3. Ridge regression (closed form, worked numerics)

Data (with intercept): $X = [[1,1],[1,2],[1,3]]$, $y = [1, 2, 2]$, $\lambda = 1$.

Compute:

$$X^T X = [[3, 6], [6, 14]]$$

$$X^T X + \lambda I = [[4, 6], [6, 15]]$$

$$X^T y = [5, 11]$$

Inverse of $\begin{bmatrix} 4 & 6 \\ 6 & 15 \end{bmatrix}$:

$$\det = 4 \cdot 15 - 6 \cdot 6 = 60 - 36 = 24$$

$$\text{inverse} = (1/24) * \begin{bmatrix} 15 & -6 \\ -6 & 4 \end{bmatrix}$$

$$\beta = (X^T X + \lambda I)^{-1} X^T y$$

$$= (1/24) * \begin{bmatrix} 15 & -6 \\ -6 & 4 \end{bmatrix} * \begin{bmatrix} 5 & 11 \end{bmatrix}^T$$

$$= (1/24) * \begin{bmatrix} 15 \cdot 5 - 6 \cdot 11 & -6 \cdot 5 + 4 \cdot 11 \end{bmatrix}$$

$$= (1/24) * \begin{bmatrix} 75 - 66 & -30 + 44 \end{bmatrix} = (1/24) * \begin{bmatrix} 9 & 14 \end{bmatrix}$$

$$= \begin{bmatrix} 0.375 & 0.583333... \end{bmatrix}.$$

Answer: intercept = 0.375, slope $\approx$ 0.583333.

4. Newton's method (nonlinear root)

Find the root of $f(x) = \cos x - x$ starting at $x_0 = 0.5$.

Derivative: $f'(x) = -\sin x - 1$.

Update: $x_{k+1} = x_k - f(x_k)/f'(x_k)$.

Iterations (rounded to 6 decimals):

$$x_0 = 0.500000$$

$$x_1 = 0.755222$$

$$x_2 = 0.739142$$

$$x_3 = 0.739085$$

$$x_4 = 0.739085 \text{ (converged)}$$

Answer: root $\approx$ 0.739085.